WEEK 6 · FINAL PROJECT

Final Project Requirements

From real NIGCOMSAT data to a working monitor your unit can use.

In one sentence: pick one real problem in **your own unit**, use **real NIGCOMSAT data** to watch it or forecast it, test your method honestly against a simple baseline, and hand your unit a working tool they can open.

This is an **individual** project. It follows the original Publica Academy capstone brief (`Time-series/module-project/`) and keeps its rubric. This document breaks that brief into simple steps, checklists and templates.

1. The journey at a glance

Stage	What you do	Skills from
0. Choose	Pick one brief + one named process in your unit. Get your unit head's OK by email	Week 2
1. Understand	Load, profile and plot your series; find its season	Weeks 1–2
2. Build	Build the method your brief needs	Weeks 3–6
3. Test honestly	Split by time, compare with a simple baseline, same scorer	Weeks 4–6
4. Deploy	A Streamlit dashboard **or** a scheduled report your unit can use	Week 6
5. Present	2–4 page report + 5-minute demo	Week 6 Day 3

Checkpoints during the course

When	Checkpoint	Bring
Week 2 · Day 3	🎯 Brief chosen	Your brief, the named process, the draft email to your unit head
Week 3 · Day 3	📥 Data in hand	Unit head's reply, your data loaded and profiled (rows, range, gaps, duplicates)
Week 4 · Day 3	🛠️ Method drafted	First version of your detector / forecast / trend-to-threshold

When	Checkpoint	Bring
Week 5 · Day 3	⚖️ Honest test	Temporal split + baseline comparison table
Week 6 · Day 3	🎤 Demo day	Working tool, report, 5-minute demo

2. Choose ONE brief

Brief	The question in plain words	Naija version	Skills
A. Telemetry anomaly detection	Which hours of subsystem telemetry are abnormal, and why?	"Which hours did the satellite behave strangely, and what was the cause?"	Weeks 1, 3, 4–5
B. Power and battery degradation	When will battery (or solar array) health cross its operational limit?	"When will the battery be too tired to carry us through eclipse?"	Weeks 1, 3, 6
C. RF link-quality prediction	Will link margin fall below threshold in the coming hours or days?	"Will the rain cut our signal tomorrow?"	Weeks 1, 4–5, 6
D. Carrier and capacity monitoring	Is utilisation or throughput drifting, and when will capacity run short?	"When will the expressway be full?"	Weeks 1, 3, 6
E. Site-downtime prediction	Which sites or regions are likely to go down, and how many tickets should the NOC expect?	"Which VSAT sites will give us wahala next week?"	Weeks 1, 4–5, 6

If your unit's process doesn't fit exactly, you may adapt a brief **with your unit head's agreement**. Say what you adapted in your report.

3. Data rules: read these twice

Rule 1: real data. Use real, accessible NIGCOMSAT data that your unit produces and you are allowed to use. The synthetic course files are for practice only (exception: your unit head confirms in writing that no real data can be made available).

Rule 2: real data stays inside NIGCOMSAT. Google Colab runs on Google's servers. **Do not upload real NIGCOMSAT data to Colab or paste it into any AI tool** unless your unit head has agreed in writing. Develop on the **training server** or your work laptop.

Rule 3: no raw data on GitHub. Commit your **code**, never the data export. Load data from a path set in a config file or `.env`.

☐ I named a **specific process** with an owner (e.g. "weekly battery-health review on the bus subsystem"), not just "satellite data".

☐ My unit head agreed **by email** to the process, the data and who the tool is for. The email is in my submission.

☐ I know which of my data is restricted, and I describe its *shape* to AI tools instead of pasting it.

☐ My repository contains no raw NIGCOMSAT data.

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4. What you must build: stage by stage

Stage 1: Understand the series

☐ Loaded with a proper `DatetimeIndex`, a declared frequency (`.asfreq()`) and **one** time zone.

☐ Profiled: rows, date range, gaps, duplicate timestamps, changes in sampling rate. I wrote what I found and what I did about it.

☐ An honest plot: sensible axis, the **operational threshold drawn on the chart**, a title that states a finding.

☐ Seasonal period found **from a correlogram** (not assumed), and used in a decomposition.

Stage 2: Build the method

Your brief	Minimum method
A. Anomaly / E. Site downtime (detection)	A statistical baseline (control chart, residual or CUSUM detector), **plus** a multivariate detector on engineered, trailing-only features if several channels are involved
B. Degradation / D. Capacity	A trend-to-threshold estimate giving a crossing date **and a window** (never a single day), with the assumptions stated
C. RF prediction / E. Ticket forecast	At least one baseline forecast (naive or seasonal naive) **plus** a regression with engineered features, **each with a prediction interval**

The leak test, for every feature: *"Could I have computed this at the moment the prediction was needed, using only what I knew then?"* If not, delete it. Trailing windows and positive shifts only.

Stage 3: Test it honestly

☐ Split **by time**, never at random. An `assert` proves every training timestamp comes before every test timestamp.

☐ An **embargo** at the split at least as long as my longest feature window.

☐ Compared with the **simple baseline**, on the **same period**, with the **same scorer**.

☐ Detection: **event-level precision and recall**, with my scoring rules written down first. **No accuracy figure.**

☐ Forecasting: a **walk-forward backtest** reporting the spread across folds, not just the average.

☐ Degradation: the call repeated from **several starting points**, showing how the date and the warning time change.

☐ Intervals: **coverage checked** on held-out data.

☐ No labelled events? I said so, and agreed a substitute (maintenance log, ticket history) with my unit.

✅ **If the simple method wins, say so and recommend it.** That is a legitimate, respected result.

Stage 4: Deploy it

Choose **one**:

Option	What it is	Must show
Streamlit dashboard	A small web app your unit opens in a browser	the series, the threshold, current flags or the forecast with its range, a one-paragraph plain-language summary
Scheduled report	A script that runs daily (cron / Task Scheduler) and saves a dated chart + summary to a shared folder	the same content, refreshed automatically

It must run on the **training server**. **Server access is rostered: at most 5 participants at a time.** Book your slot, build and test locally first, and use the server for deployment and final checks. A tool that runs only on your laptop earns at most half of the deployment marks.

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5. What you hand in

☐ **Repository** (or folder): code, `requirements.txt`, `README.md` explaining how to run it. **No raw data.**

☐ **The running tool** on the training server: its URL or schedule written in the README.

☐ **Report** (2–4 pages, Markdown or PDF): unit, process, brief and the unit head's agreement; the data (source, span, cadence, quality issues and fixes); the method and why it fits; the evaluation table against the baseline, with rules stated; a **4-sentence recommendation**; limitations.

☐ **AI-use note** (one paragraph): what you used AI for, and how you checked it without exposing restricted data.

☐ **5-minute demo** to the cohort and, if possible, your unit head.

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6. How it's graded (100 points)

Criterion	What earns the points	Points
Problem framing and data	Unit, named process, one brief, unit head's agreement; real data described (source, span, cadence); restrictions respected; no raw data committed	**15**
Understanding the series	DatetimeIndex, declared frequency, one time zone; gaps and duplicates handled; honest plot with threshold; period from a correlogram + decomposition	**15**

Criterion	What earns the points	Points
Method	Suits the brief; every feature passes the leak test	**20**
Honest evaluation	Temporal split with assert + embargo; same period, same scorer vs the baseline; spread / repeated calls / coverage reported; conclusion follows the evidence	**25**
Deployment	Dashboard or scheduled report running on the training server; README explains how to run it	**15**
Communication and AI use	4-sentence recommendation, limitations stated, readable by the unit head; AI-use note; demo delivered	**10**

Automatic penalties: a random train/test split · any feature that uses future information · an accuracy figure for rare-event detection · a degradation estimate given as a single date with no window or assumptions · **committing restricted data** (treated as a serious failure and reported to the programme lead).

Not penalised: a model that loses to the simple baseline, as long as the comparison is fair and you recommend the simpler method.

7. Templates you can copy

Email to your unit head

Subject: Time-series training: final-project scope for approval

Dear [Name],

For my final project in the NIGCOMSAT time-series programme, I'd like to build a small monitoring tool for **[named process, e.g. the weekly battery-health review]**. - **Brief:** [A–E] - **Data I'd use:** [system/export], [date range], [cadence]. It stays on NIGCOMSAT systems; nothing is uploaded to outside tools. - **Who it's for:** [team / role] - **Output:** a [dashboard / daily report] on the training server.

Could you confirm by replying to this email? Thank you.

The 4-sentence recommendation

1. **Situation:** what the unit does today, and the problem.
2. **Observation:** what the data and your method showed (one number).
3. **Implication:** what it means for operations.
4. **Recommendation:** what the unit should do next, and with which tool.

Limitations: answer these three

- **What will it miss?** (blind spots)
- **How late will it be?** (detection delay / forecast horizon)
- **What did you assume was normal?** (the training period)

8. Hardware and data-collection recommendations

Good models need good data. These recommendations are for **NIGCOMSAT units and IT/infrastructure teams** that want better data for this kind of work. Treat them as a menu: agree priorities and budget with your unit; some may already be in place.

8.1 For the course itself

Item	Recommendation	Why
Participant laptops	Any laptop from the last ~6 years with **8 GB RAM** and **Google Chrome**	Colab does the computing; the laptop only shows the screen
Internet	Stable broadband, at least **~2 Mbps per participant**, plus a backup connection	Colab needs a live connection; disconnects lose memory
Power	UPS or inverter for the training room, projector and router	A NEPA cut shouldn't end a session 😅
Projector	1080p, HDMI; a clicker for the animated slides	The decks are designed for 16:9

8.2 The training server (for deployment)

Item	Recommended minimum	Why
CPU / RAM	**16 vCPU, 64 GB RAM**	5 concurrent users × (2 vCPU + 8 GB) plus headroom for dashboards
Storage	**1 TB SSD** (+ backup)	Years of hourly telemetry for many channels, plus model outputs
GPU	**Not needed**	Every method in this course is CPU-light
Software	Linux (Ubuntu LTS), Python 3.11+, JupyterHub, Streamlit, cron, Git	Development, dashboards and scheduled reports on one box
Access	Named user accounts, a booking roster, VPN for remote access	The rostered 5-user limit, plus security
Power	UPS + generator changeover	Scheduled reports must run every day

8.3 Better data at the source: what to collect, and how

Area	Recommendation	What it unlocks
Time	**NTP or GPS-disciplined time sync** on every logger, server and modem. Store all timestamps in **UTC**	Events from different systems line up; no hour-shift joins
Telemetry archive	Keep **full-rate** satellite telemetry (not just daily summaries) in a time-series database (e.g. TimescaleDB, InfluxDB), and keep **5+ years**	Briefs A and B: degradation forecasting needs long history

Area	Recommendation	What it unlocks
Weather at earth stations	A **rain gauge / weather station** logging at 1-minute resolution beside each Ka/Ku-band antenna	Brief C: separates **rain fade** from **equipment faults**, which gives you free labels
Propagation	A **Ka-band beacon receiver** measuring signal attenuation directly	Brief C: a clean, physical measure of rain impact
Carrier monitoring	A carrier-monitoring system / spectrum analyser logging **C/N, Eb/N0, EIRP** at fixed intervals	Briefs C and D: link quality and capacity trends
VSAT sites	**SNMP polling** of modem, BUC and LNB metrics every 1–5 minutes into the time-series database	Brief E: early warning before a site drops
Site power	**Power monitoring** at remote sites: grid vs generator vs battery, voltage, changeovers (smart meter / ATS logs)	Brief E: in Nigeria, power is a top cause of site downtime. Now you can prove it
Store and forward	Local buffering on site loggers so data isn't lost during link outages	No gaps exactly when things go wrong
NOC tickets	Structured fields: **site ID, cause code, open/close time (UTC), linked alarm ID**	Brief E: ticket counts become trustworthy labels
Maintenance log	A **machine-readable** log of interventions, faults and fixes (date, asset, what happened)	Briefs A and B: ground truth for scoring detectors
Data contract	Fixed export schema + data dictionary (column names with units, ISO dates, empty = missing), plus an automatic daily `quick_check` for gaps and duplicates	Every brief: no silent data problems
Security	Role-based access, audit logs, a clear list of what is restricted	Safe use of AI tools; compliance

💡 **Quick wins (low cost, big impact):** time sync to UTC, a rain gauge at each Ka-band station, structured NOC cause codes, and a machine-readable maintenance log.

9. FAQ

Can I use the course's synthetic data? For practice, yes. For the submission, only if your unit head confirms in writing that no real data could be made available, and you include that email.

My model is worse than the baseline. Did I fail? No. Recommend the baseline, show the fair comparison, explain why. That's full marks for honesty.

Can I work in Colab? Yes, with the synthetic data while you learn. Real data stays on NIGCOMSAT systems unless your unit head agrees otherwise in writing.

Can I use AI to write code? Yes, but you must understand and verify every line, never paste restricted data, and describe how you used it in your AI-use note.

We have no labelled faults. Agree a substitute with your unit (maintenance log, ticket history, operator notes) and say so in the report.